Minseok Park

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Education

Hanyang University | Seoul, Republic of Korea
B. S. Electronic Engineering | Expected Feb 2027
- GPA: 4.21/4.5

University of North Dakota | Grand Forks, ND
Exchange Program, Electrical Engineering | Jan - May 2026
- GPA: 4.0/4.0 | Advisor: Prof. Jueming Hu

Publication

Park, M., Yang, C. X., Hu, J. (2026). Robust fault diagnosis of rotating machinery using entropy-augmented CNN in time and frequency domains. Accepted for the ASME International Mechanical Engineering Congress and Exposition (IMECE)

Research Experience

Resilient UAV Navigation under Sensor Dropout
University of North Dakota | Jan 2026 - Present
- Developed a UAV navigation policy resilient to sensor dropout, achieving up to 91– 95% task completion rate and outperforming baseline models under intermittent sensor failure conditions. Manuscript in preparation, targeting IEEE RA-L.

Robust Fault Diagnosis of Rotating Machinery via Entropy-Augmented CNN
University of North Dakota | Jan 2026 - May 2026
- Developed a fault detection framework using entropy-based signal processing and CNN classification to identify mechanical imbalance and misalignment, achieving over 95% classification accuracy across varying rotational speeds. Published at ASME IMECE 2026.

End-to-End RL-Based UAV Waypoint Navigation
University of North Dakota | Jan 2026 - May 2026
- Developed an end-to-end RL-based navigation controller simulating Crazyflie UAV dynamics in PyBullet, implementing single and multi-waypoint goal tracking.

Multi-Anchor CNN with ToA Attention for UWB Indoor Localization
Hanyang University | Sep 2025 - Dec 2025
- Developed a Multi-Anchor CNN model integrating ToA-based attention to estimate receiver position for UWB-based indoor localization.

Technical Skills

Programming Languages: Python
AI & Machine Learning: PyTorch, CNN, Reinforcement Learning (RL), Imitation Learning (IL)
Robotics & Simulation: PyBullet, Crazyflie simulation